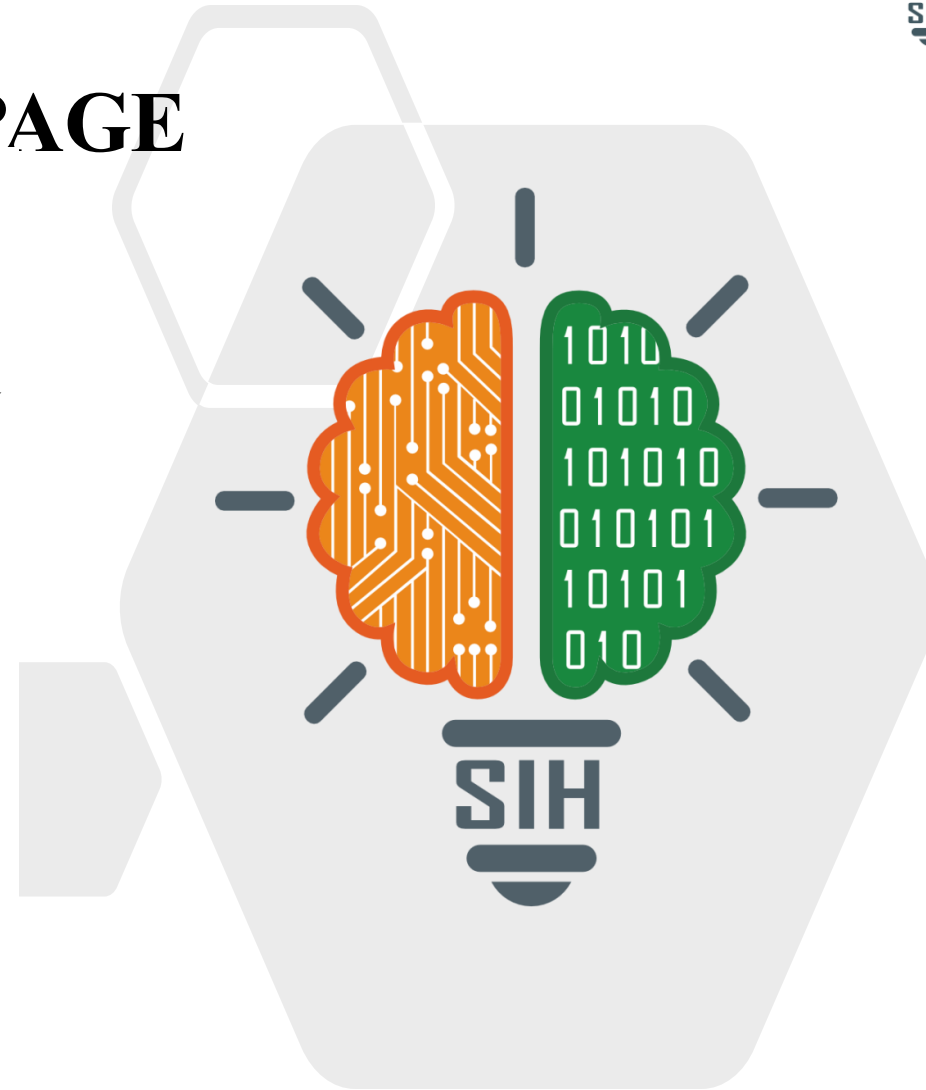


TITLE PAGE

- Problem Statement ID – SIH26153
- Problem Statement Title- *"AI Based Network Attack Forecasting Using Network Traffic Data"*
- Theme- Blockchain & Cybersecurity
- PS Category- Software
- Team ID- SIH-A0H-T379
- Team Name (Registered on portal) - TechTonic



AI-Based Network Attack Forecasting Proposed Solution

ONGOING CHALLENGES

- ! **Detected too late** — attacks are flagged only after damage is done
- ! **Evasion** — slow, low-and-slow attacks slip past flow-based thresholds
- ! **Alert fatigue** — constant false positives from normal traffic
- ! **No calibration** — confidence scores don't mean what they claim
- ! **Model drift** — network behaviour changes, the model doesn't
- ! **No fallback** — detector has no backup when overwhelmed
- ! **Black-box alerts** — analysts can't act on unexplained scores

OUR SOLUTION

A calibrated, explainable, fast + slow dual-path forecasting system that predicts attacker progression across MITRE ATT&CK stages before compromise completes.

FUSION MODEL

- ✓ LSTM / Transformer for temporal patterns
- ✓ GraphSAGE for structural patterns

CONTINUOUS MONITORING

- ✓ 15-min window forecasts the next 10 min
- ✓ Explainability across the ATT&CK pipeline

RELIABILITY LAYERS

- ✓ Per-stage calibration + drift handling
- ✓ Overload fallback, compression & serving

ATTACK PROGRESSION

- ✓ Probability of attack at every stage
- ✓ Predicts how current behaviour evolves

Calibrated

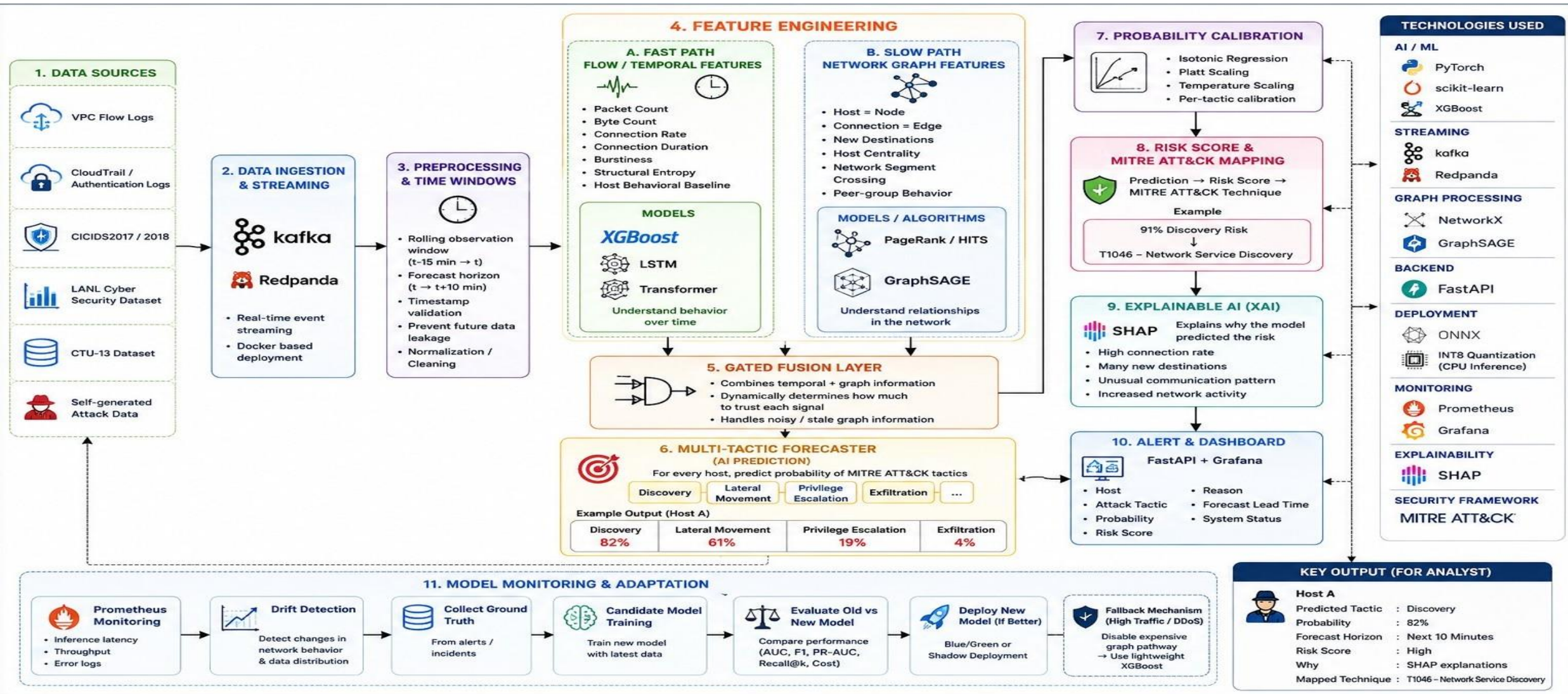
Explainable

Fast + Slow Path

Drift-Aware

AI-Based Network Attack Forecasting

Technical Approach



AI-Based Network Attack Forecasting Feasibility & Viability

TECHNICAL

- ✓ Well-documented open-source tech (LSTM/Transformers)
- ✓ Publicly available labelled datasets
- ✓ Model ladder: Logistic Regression -> XGBoost -> LSTM -> GraphSAGE
- ✓ Fast/slow dual-path architecture
- ✓ Model compression + built-in safety checks

FINANCIAL

- ✓ Free open-source tools: Kafka, PyTorch, etc.
- ✓ Low hardware need — runs on CPU
- ✓ Reduced operational cost
- ✓ AWS free-tier for cloud experimentation
- ✓ Free / public datasets

OPERATIONAL

- ✓ Real-time continuous monitoring
- ✓ Human-readable risk + stage probabilities
- ✓ Adaptive to changing networks
- ✓ Overload fallback keeps it reliable

SCALABILITY

1

Runs fully on-premises / offline — secure by design

2

Uses traffic metadata & flow stats only — no payload inspection

3

Modular architecture scales horizontally

4

Dynamic peer re-clustering + per-tactic calibration adapts to new environments

AI-Based Network Attack Forecasting

Impact & Benefits

IMPACT

- ▲ Reduced cybersecurity risk exposure
- ▲ Faster incident response
- ▲ Protection of critical infrastructure
- ▲ Reduced analyst workload
- ▲ Improved security decision-making
- ▲ Greater resilience against evolving attacks
- ▲ Lower cost of cyber defence

BENEFITS

- ✓ **Proactive defence** — early warning time to investigate and respond
- ✓ **Fewer false alerts** — learns normal per-host behaviour to separate genuine from noise
- ✓ **Explainable** — every prediction comes with a SHAP-based reason
- ✓ **Adaptive** — keeps pace with changing network behaviour
- ✓ **Cost-efficient** — open-source stack, CPU-friendly, low deployment cost

AI-Based Network Attack Forecasting Research & References

RESEARCH BASIS

Hybrid ARIMA + Decision Trees on MITRE ATT&CK-Labeled Zeek Logs

- ARIMA identifies temporal anomalies in traffic
- Improves detection of stealthy, low-volume attacks
- Strong gains for Discovery, Persistence & Defense Evasion
- Confirms the value of time-based network behaviour

Reference [1]

Deep Learning Frameworks for DDoS Forecasting with Neurosymbolic Reasoning

- Temporal encoding profiles real-time traffic behaviour
- Multimodal transformer fuses threat-related features
- Concept-drift adaptation tracks changing attack patterns
- Explainable threat inference for interpretable predictions

Reference [2]

REFERENCES

- [1] Freeman et al., A Hybrid Time Series Forecasting Model Combining ARIMA and Decision Trees..., Electronics, 2026.
- [2] CICIDS2017/2018 Dataset — Canadian Institute for Cybersecurity.
- [3] LANL Cyber Security Dataset.
- [4] CTU-13 Botnet Dataset.
- [5] MITRE ATT&CK Framework.

LINKS

- ▶ <https://www.mdpi.com/2079-9292/15/4/871>
- ▶ <https://ieeexplore.ieee.org/abstract/document/11441009>
- ▶ CICIDS Dataset - <https://www.unb.ca/cic/datasets/ids-2017.html>
- ▶ LANL Dataset - <https://csr.lanl.gov/data/>
- ▶ MITRE ATT&CK - <https://attack.mitre.org/>